LIPIDS CHEMISTRY

DEFINITION

- HETEROGENOUS group of compounds related more physically than chemically.

(1) relatively **insoluble in water**

(2) soluble in non polar solvents such as ether and chloroform

- fats, oils, steroids, waxes, and related compounds

Importance of lipids

1.High energy value

2.Contain fat-soluble vitamins and other lipophilic micronutrients.

fat soluble vitamins: A,D,E,K

3.Contain essential fatty acids

4.Thermal insulators

5. Electrical insulators: myelinated & non myelinated nerve fibres (myelin sheath).

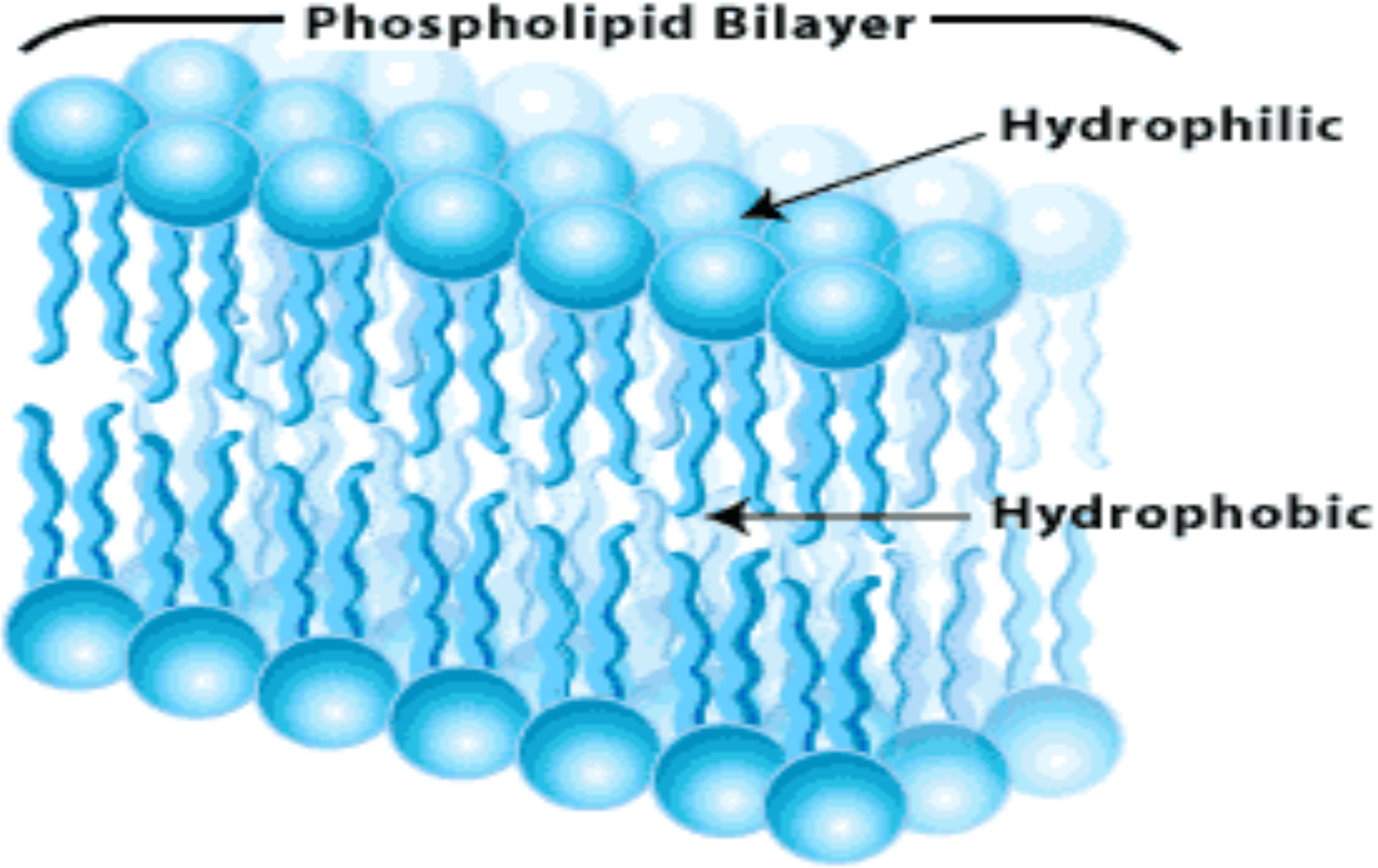
Importance of lipids

6.source of 2nd messengers

7.Constituent of cell membrane

8.Related to diseases:

- Obesity, diabetes mellitus
- Atherosclerosis: deposition of cholesterol in blood vessels → narrowing, rigidity,↓ tissue blood flow (ischemia)


Phospholipid Bilayer
Hydrophilic
Hydrophobic

Classification of lipids

- **Simple**
- **Complex**
- **Precursor & derived lipids**

Simple lipids

- These are esters of fatty acids with various alcohols:

A. Fats: esters of fatty acids with GLYCEROL. (oil is a fat in the liquid state)

TAG: triacylglycerols

b. Waxes: Esters of fatty acids with higher molecular weight monohydric alcohol

Complex lipids

Esters of fatty acids+ alcohol+ other group:

Complex lipids

a. Phospholipids: contain fatty acids and an alcohol, a phosphoric acid residue. phosphoric acid

b. Glycolipids (glycosphingolipids): contain a fatty acid, sphingosine, and carbohydrate

c. Other complex lipids: Lipids such as sulfolipids and amino lipids. Lipoproteins may also be placed in this category.

Precursor & derived lipids

These include fatty acids,
glycerol, steroids, other alcohols, fatty aldehydes, ketone bodies , hydrocarbons, lipid-soluble vitamins and micronutrients, Cholesterol and hormones

Fatty acids

- are aliphatic carboxylic acids
- In the body can be found as free fatty acids or esters in fats and oils
- FA in fats usually contain an even number of carbon atoms
- The chain may be saturated (containing no double bonds) or unsaturated (containing one or more double bonds)

Fatty acids

- Are named after the hydrocarbon with the same number and arrangement of carbon atoms with –oic

- saturated acids end in -anoic, for example, octanoic acid (C8)

- unsaturated acids with double bonds end in -enoic, for example, octadecenoic acid (oleic acid, C18).

Fatty acids

Numbering of carbon atoms:

Carboxyl carbon →1

Alpha carbon (α) →2

Beta carbon (β) →3

Gamma carbon(γ) → 4

The terminal methyl(CH_3) →(ω) or (n) carbon atom.

Fatty acids

18 17 16 15 14 13 12 11 10 9 1 18:1;9 or Δ9 18:1

$$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH{=}CH(CH_2)_7COOH$$

ω or n-1 2 3 4 5 6 7 8 9 10 18

Fatty acids

Unsaturated fatty acids

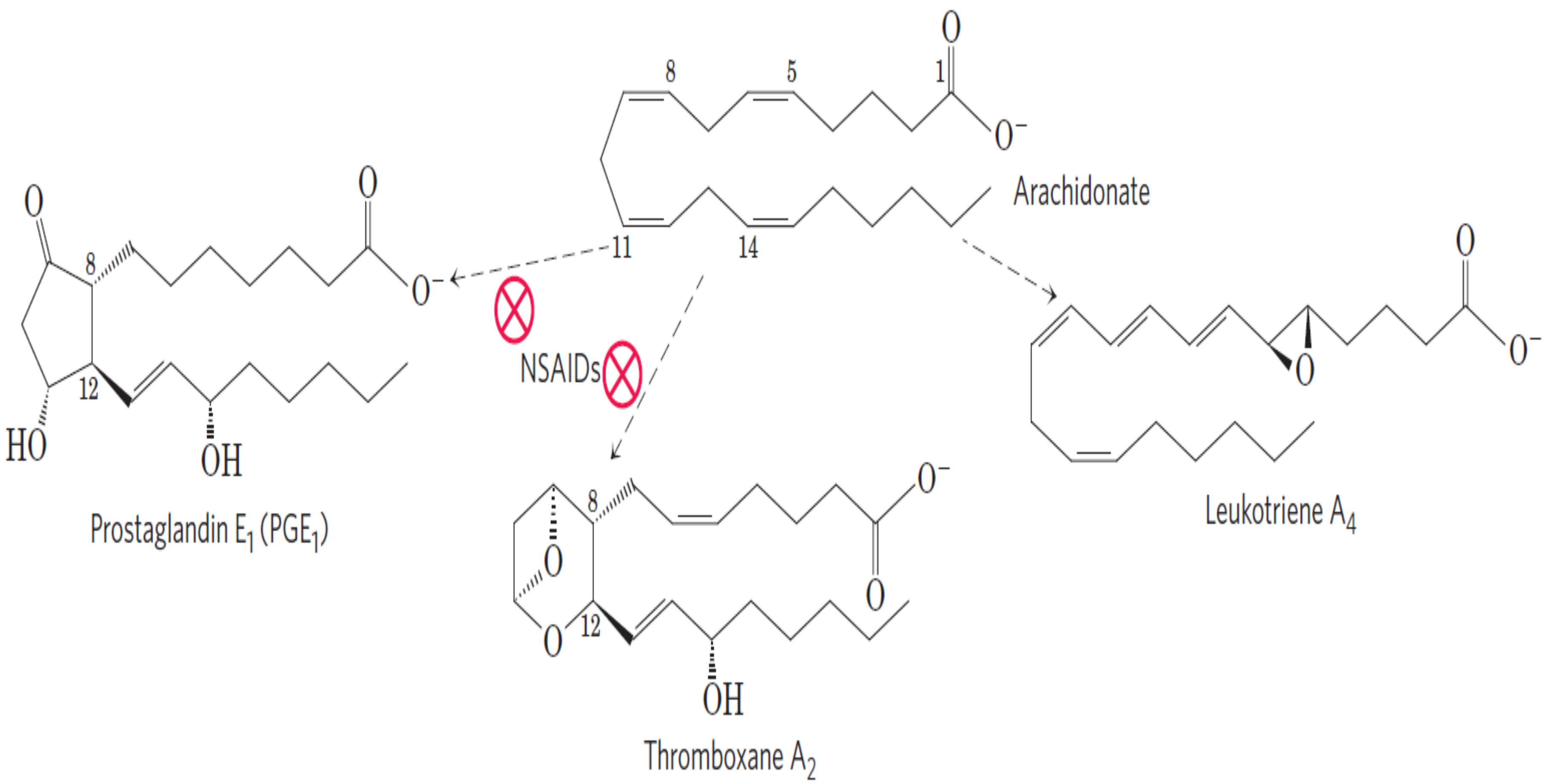
1. Monounsaturated (monoethenoid, monoenoic) acids, containing one double bond.

2. Polyunsaturated (polyethenoid, polyenoic) acids, containing two or more double bonds.

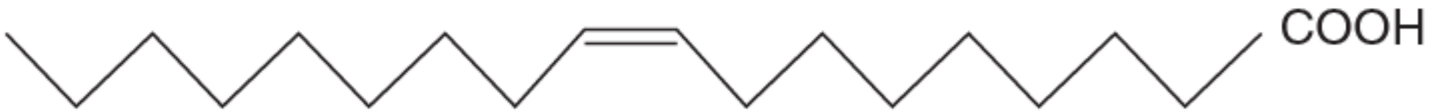
3. Eicosanoids: These compounds, derived from eicosa (20-carbon) polyenoic fatty acids

Eicosanoids

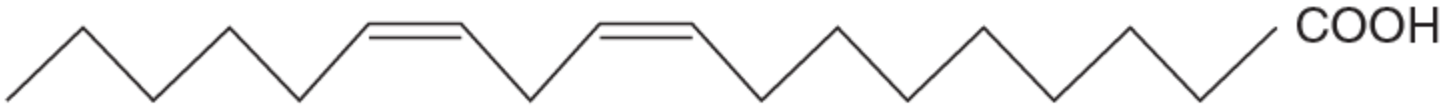
- are formed from 20-carbon polyunsaturated fatty acids and make up an important group of physiologically and pharmacologically active compounds known as prostaglandins, thromboxanes, leukotrienes, and lipoxins.


Arachidonate
Prostaglandin E1 (PGE1)
NSAIDs
Thromboxane A2
Leukotriene A4

A saturated fatty acid (palmitic acid, C16)

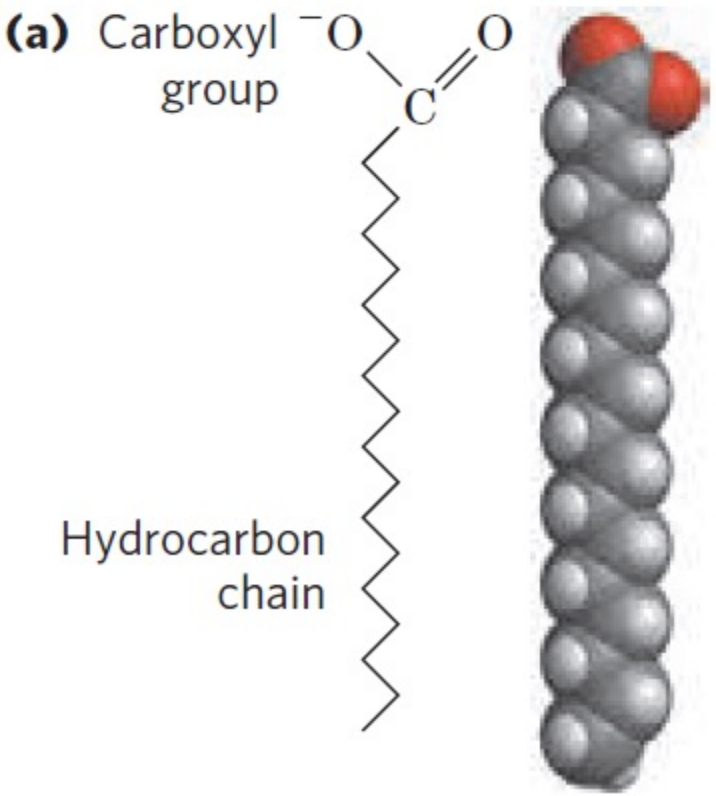


A monounsaturated fatty acid (oleic acid, C18:1)



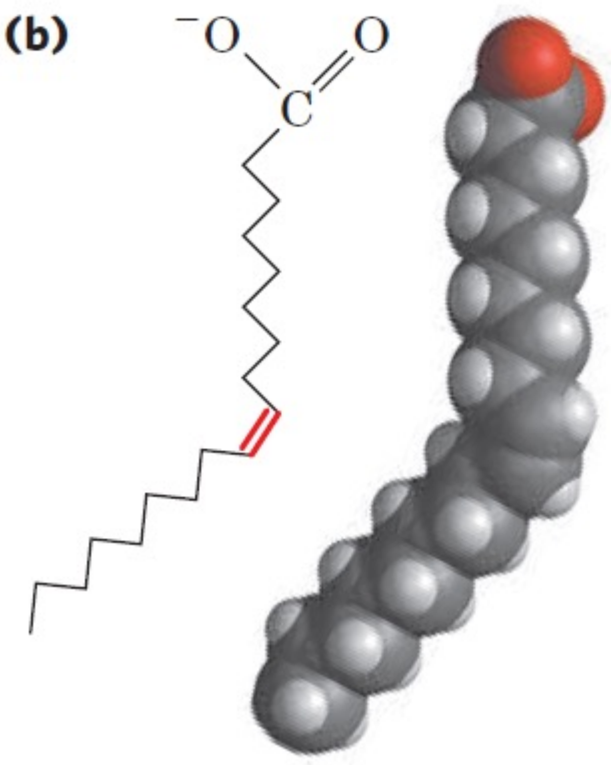
A polyunsaturated fatty acid (linoleic acid, C18:2)

Saturated Fatty acids



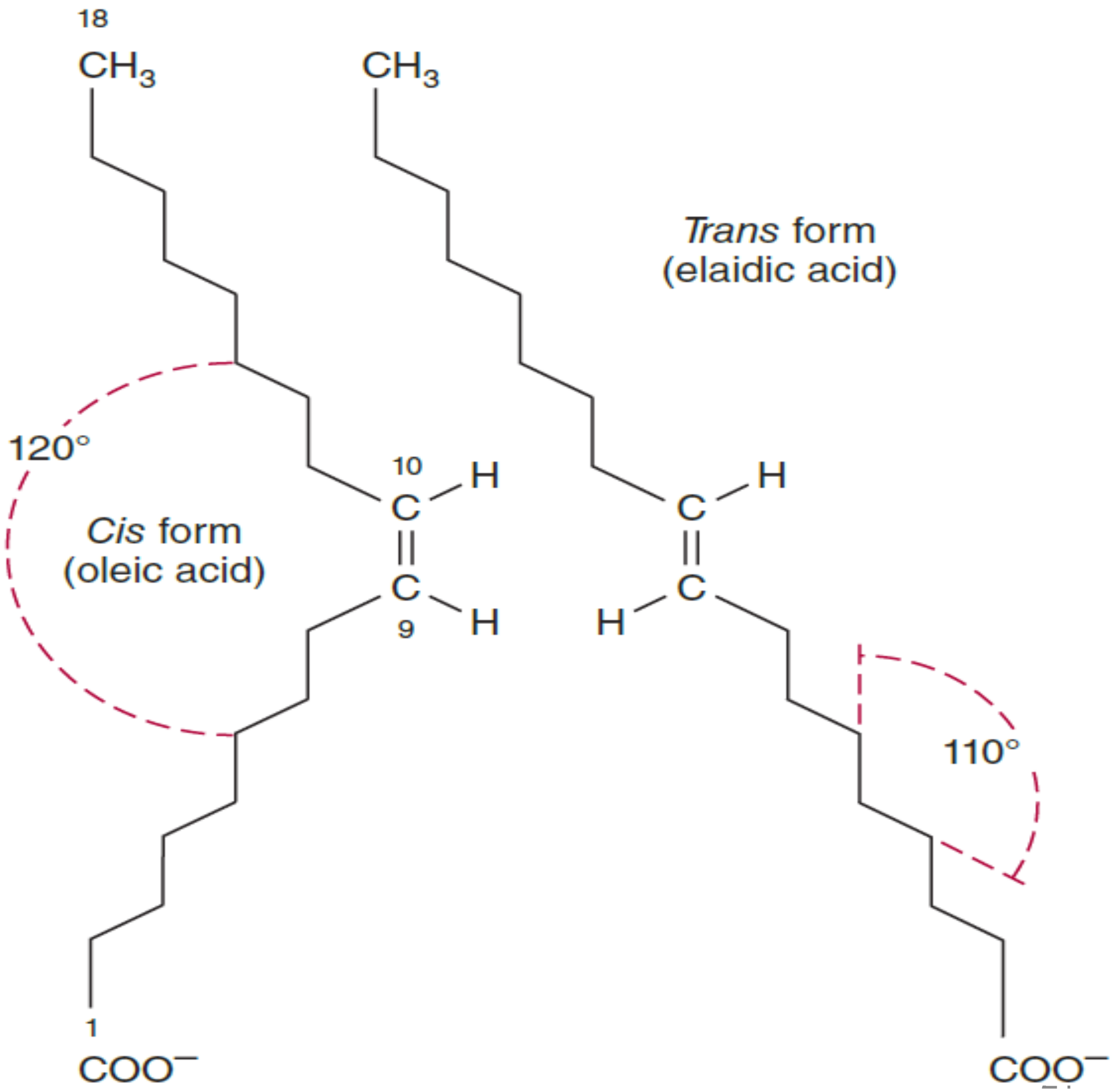
Mono unsaturated Fatty acids

(b)

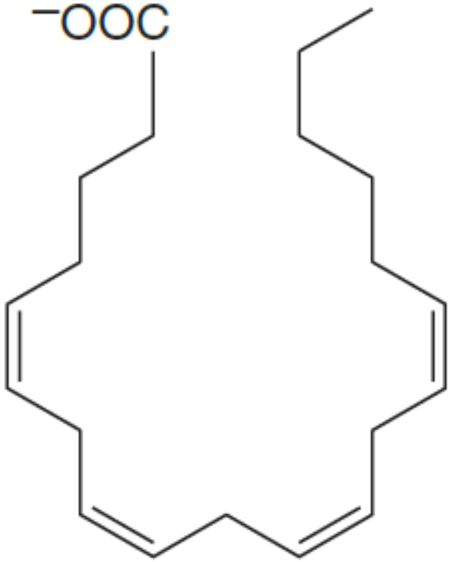


Geometric isomerism

- **of Δ9, 18:1 fatty acids (oleic and elaidic acids)**



Arachidonic acid



Fatty acids

- **Properties**
  - **Saturated fatty acids are solid at room temperature and have a high melting point**
  - **Unsaturated fatty acids are liquid at room temperature and have a low melting point**

Fatty acids

- Long chain ω3 fatty acids such as `-linolenic (ALA) (found in plant oils), eicosapentaenoic (EPA) (found in fish oil) and docosahexaenoic (DHA) (found in fish and algal oils) have anti-inflammatory effects

- Current evidence suggests that diets rich in ω3 fatty acids are beneficial, particularly for **cardiovascular disease, but also for other** chronic degenerative diseases such as **cancer, rheumatoid arthritis, and Alzheimer disease.**

Simple lipids

- These are esters of fatty acids with various alcohols:

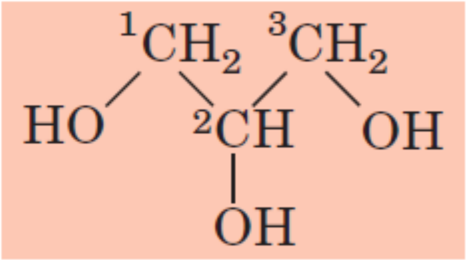
A. Fats: esters of fatty acids with GLYCEROL. (oil is a fat in the liquid state)

TAG: triacylglycerols

b. Waxes: Esters of fatty acids with higher molecular weight monohydric alcohol

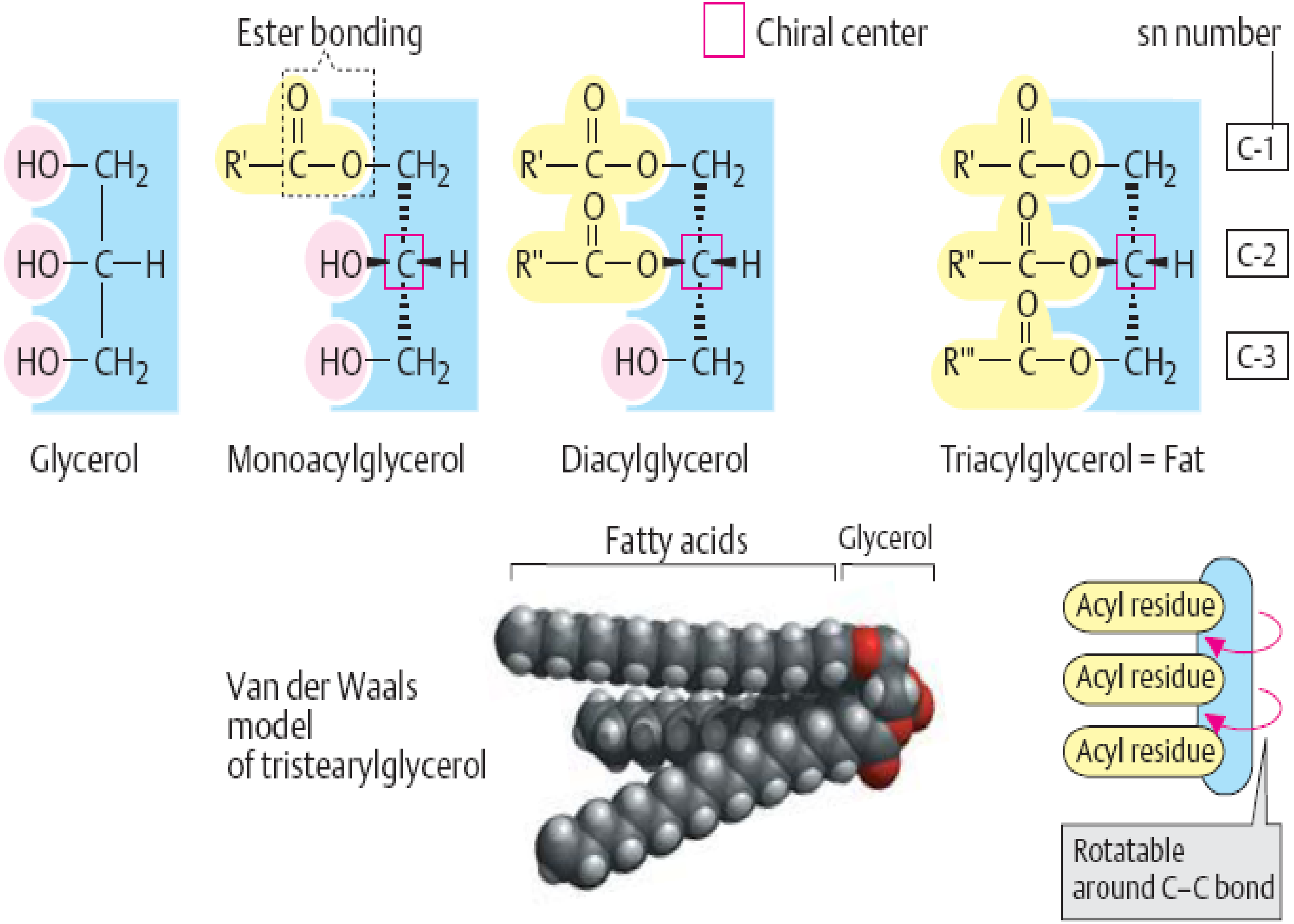
TRIACYLGLYCEROLS (TRIGLYCERIDES)

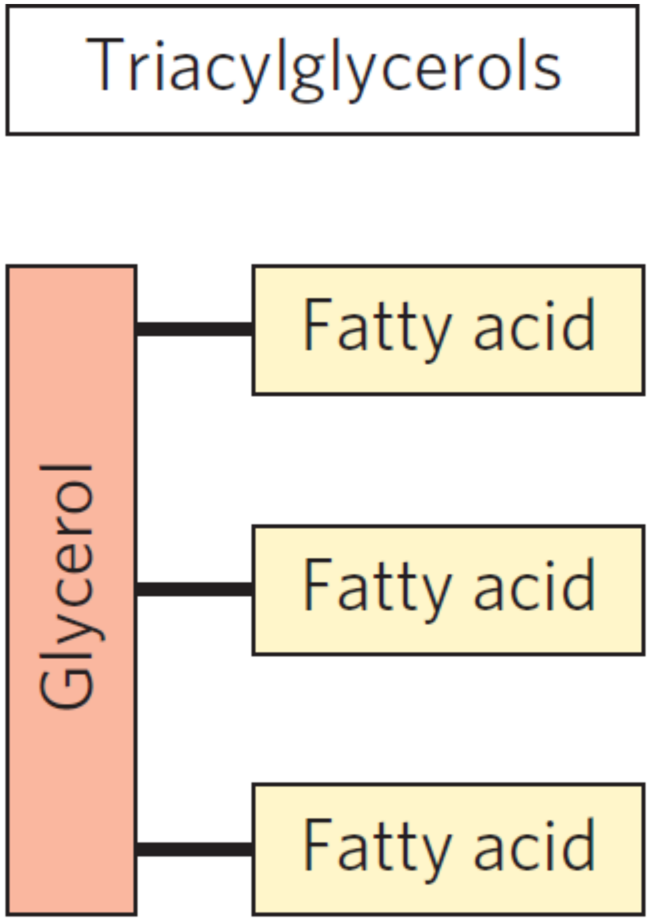
- Main storage form of fatty acids in adipose tissue.
- Esters of fatty acids with glycerol

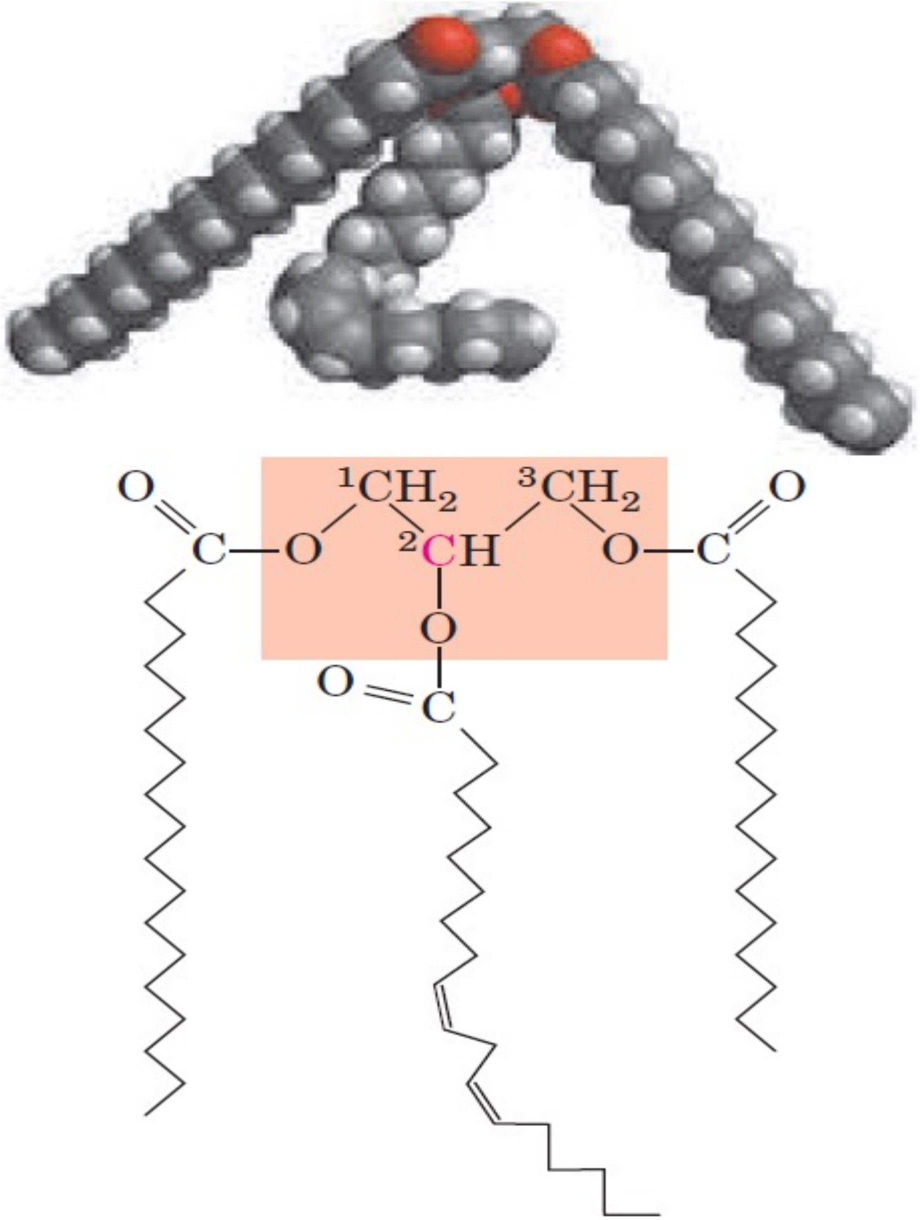


Glycerol

B. Structure of fats




Triacylglycerols
Glycerol
Fatty acid
Fatty acid
Fatty acid



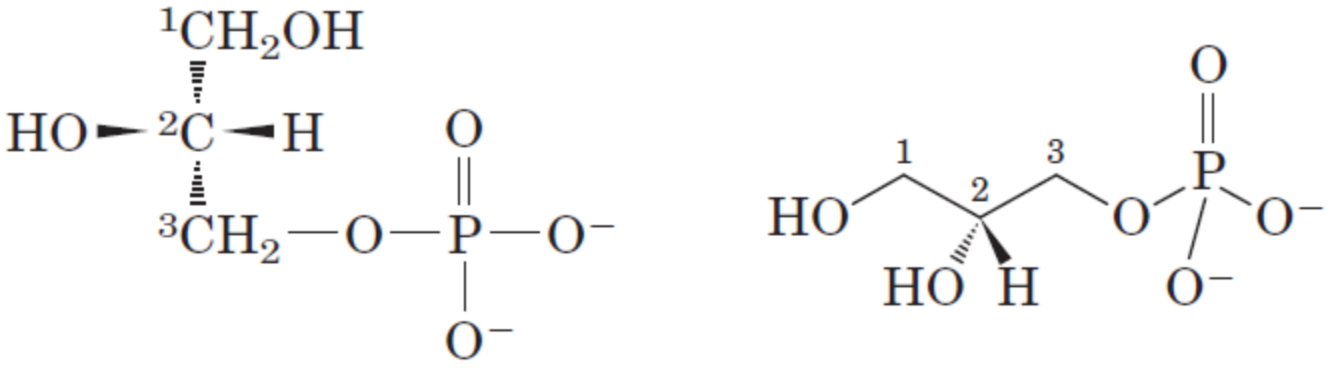
1-Stearoyl, 2-linoleoyl, 3-palmitoyl glycerol,
a mixed triacylglycerol

Complex lipids

Esters of fatty acids+ alcohol+ other group:

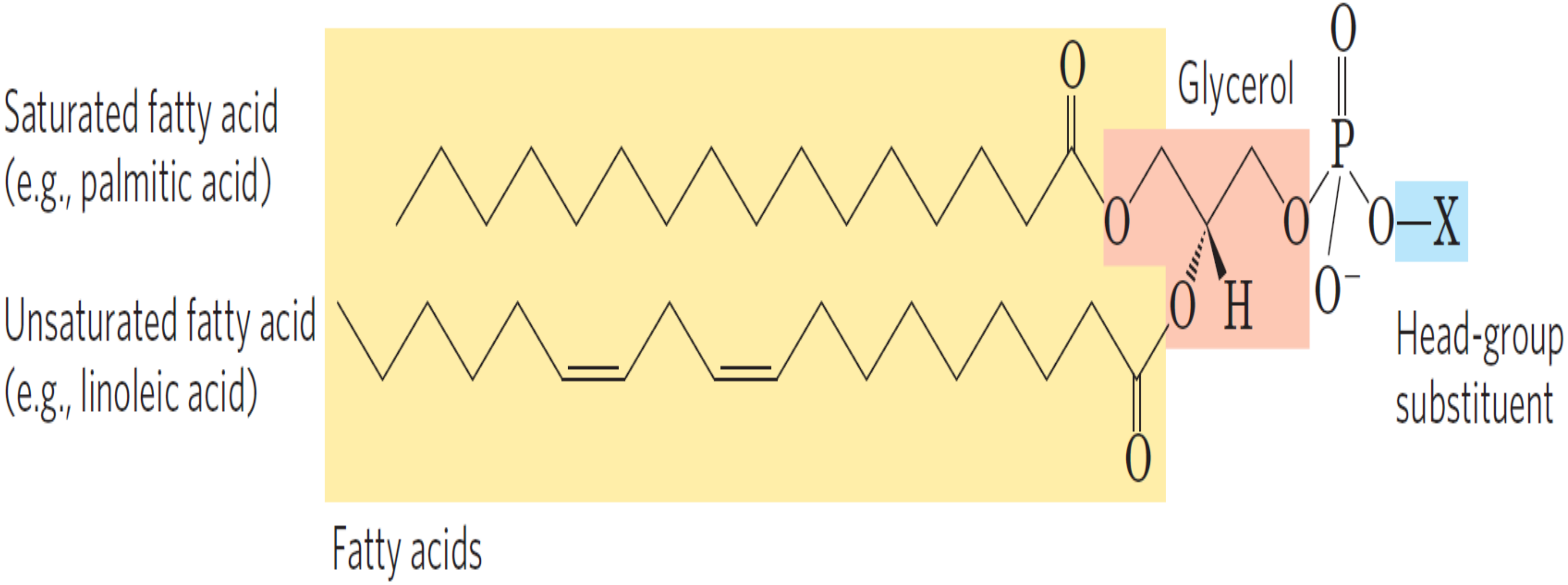
PHOSPHOLIPIDS

- Are the main lipids in membranes

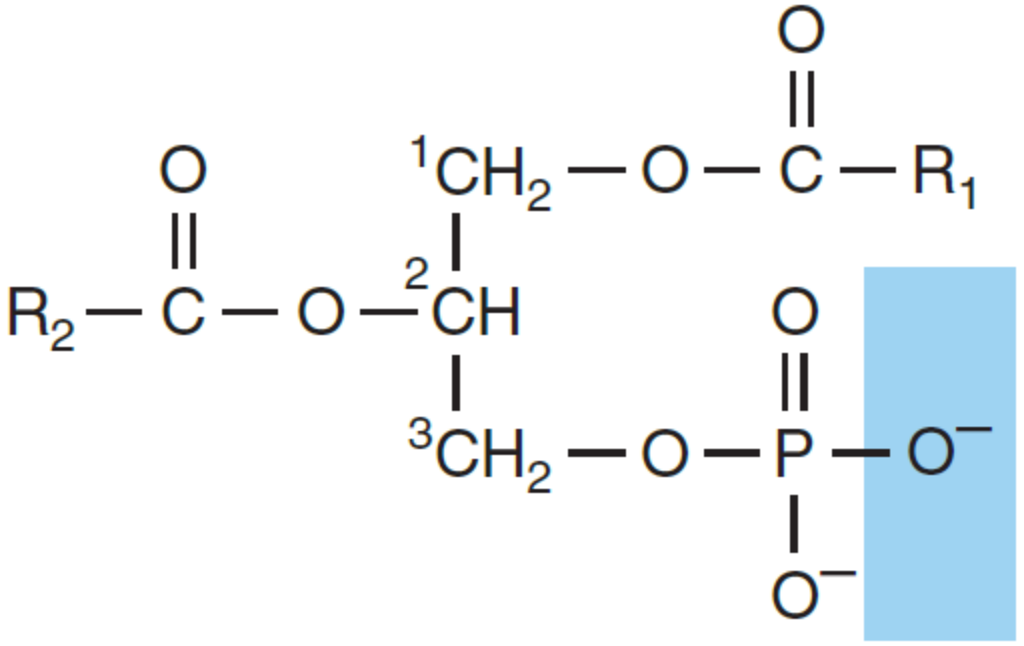


L-Glycerol 3-phosphate (*sn*-glycerol 3-phosphate)

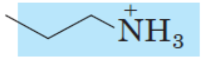
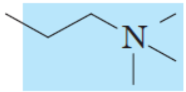
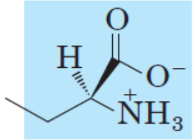
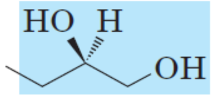
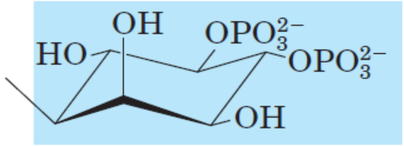
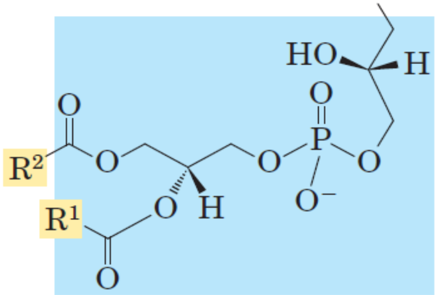
PHOSPHOLIPIDS



PHOSPHOLIPIDS

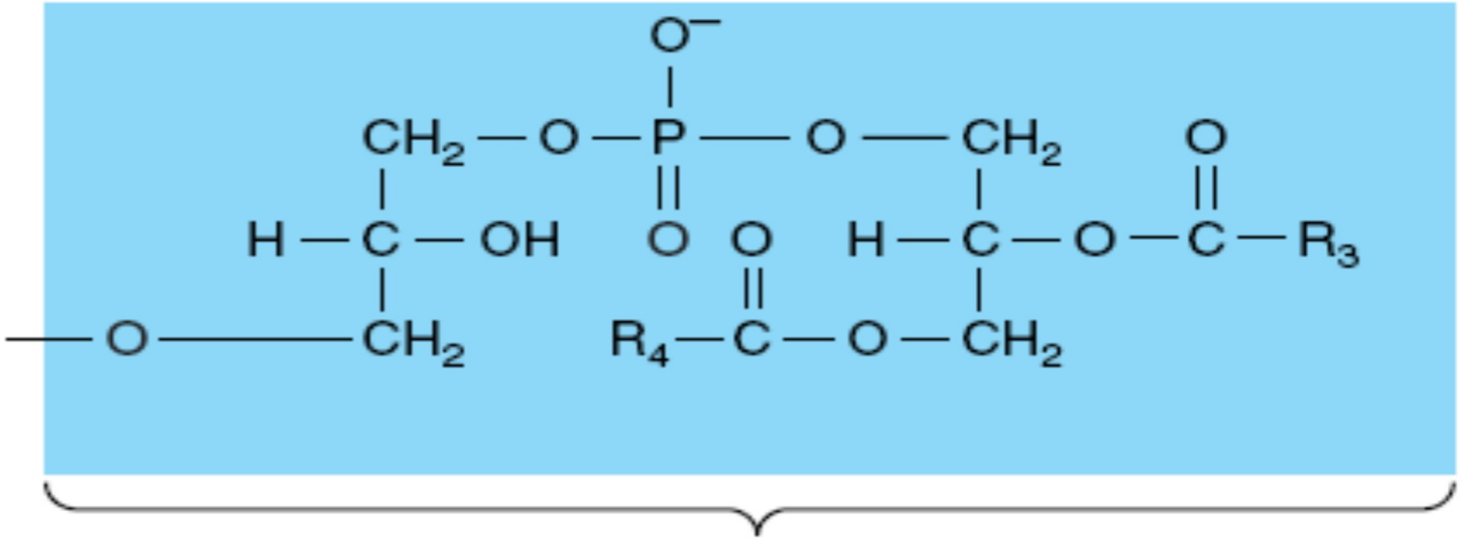


Phosphatidic acid

Name of glycerophospholipid	Name of X—O	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—	— H	−2
Phosphatidylethanolamine	Ethanolamine		0
Phosphatidylcholine	Choline		0
Phosphatidylserine	Serine		−1
Phosphatidylglycerol	Glycerol		−1
Phosphatidylinositol 4,5-bisphosphate	*myo*-Inositol 4,5-bisphosphate		−4*
Cardiolipin	Phosphatidyl-glycerol		−2

Phosphatidylglycerol

- **Phosphatidic acid + glycerol**
- **Found in the lung and aminiotic fluid indicate lung maturity**

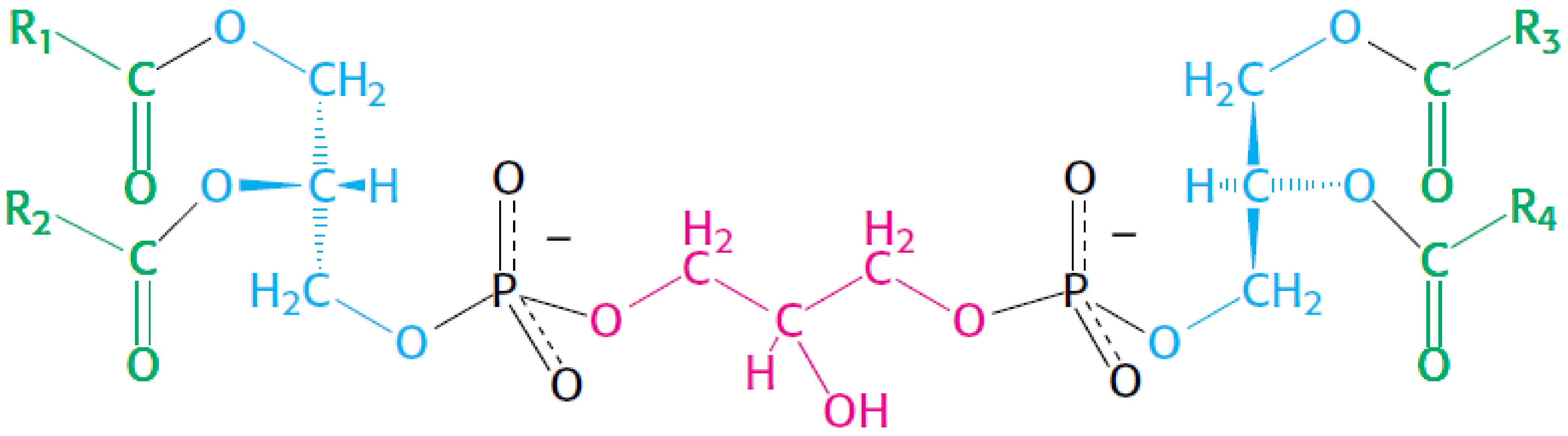


Phosphatidylglycerol

Cardiolipin

Diphosphatidyl glycerol → 2 phosphatidic acid + glycerol

Major lipid of mitochondrial membrane

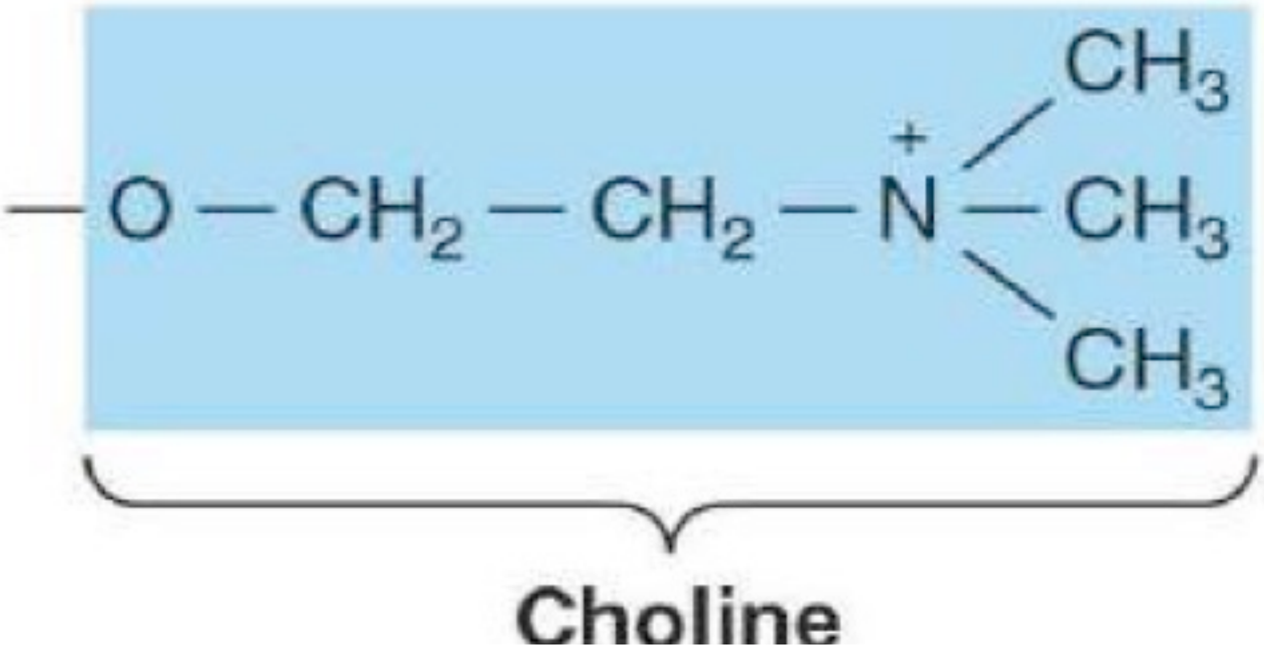


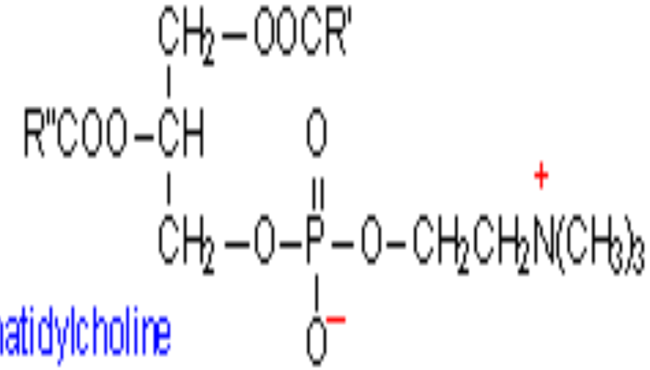
Diphosphatidylglycerol (cardiolipin)

Phosphatidylcholine

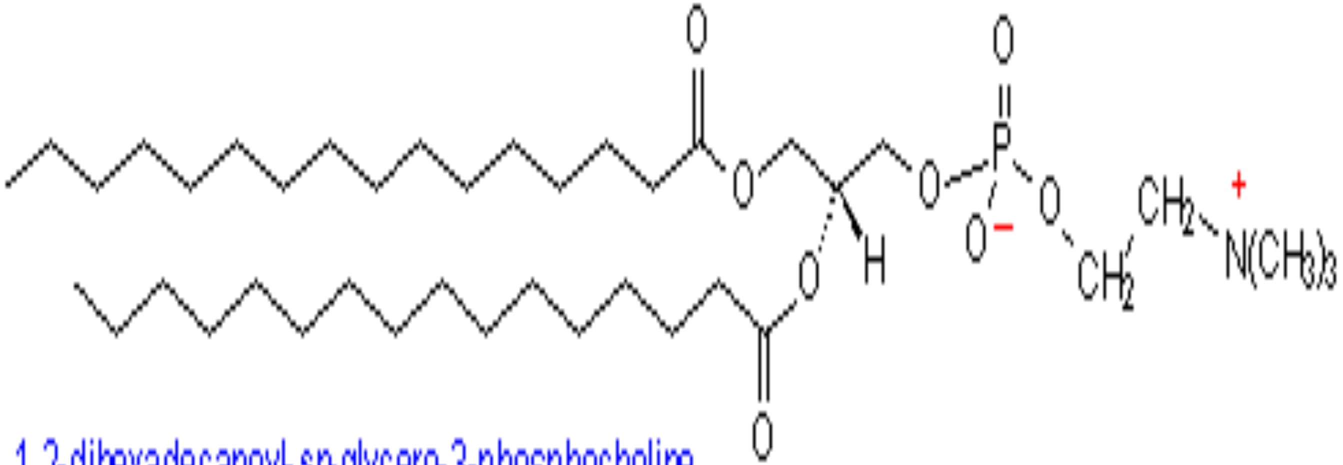
- **Consist of:**

  Phosphatidic acid + choline
- **Choline : is water soluble vitamin like nutrient.**
- **Mobile storage for methyl group**
- **(CH3)3 N CH2CH2 OH.**
- **Precursor for acetyl choline.**
- **Also found in phosphtidylcholine.**

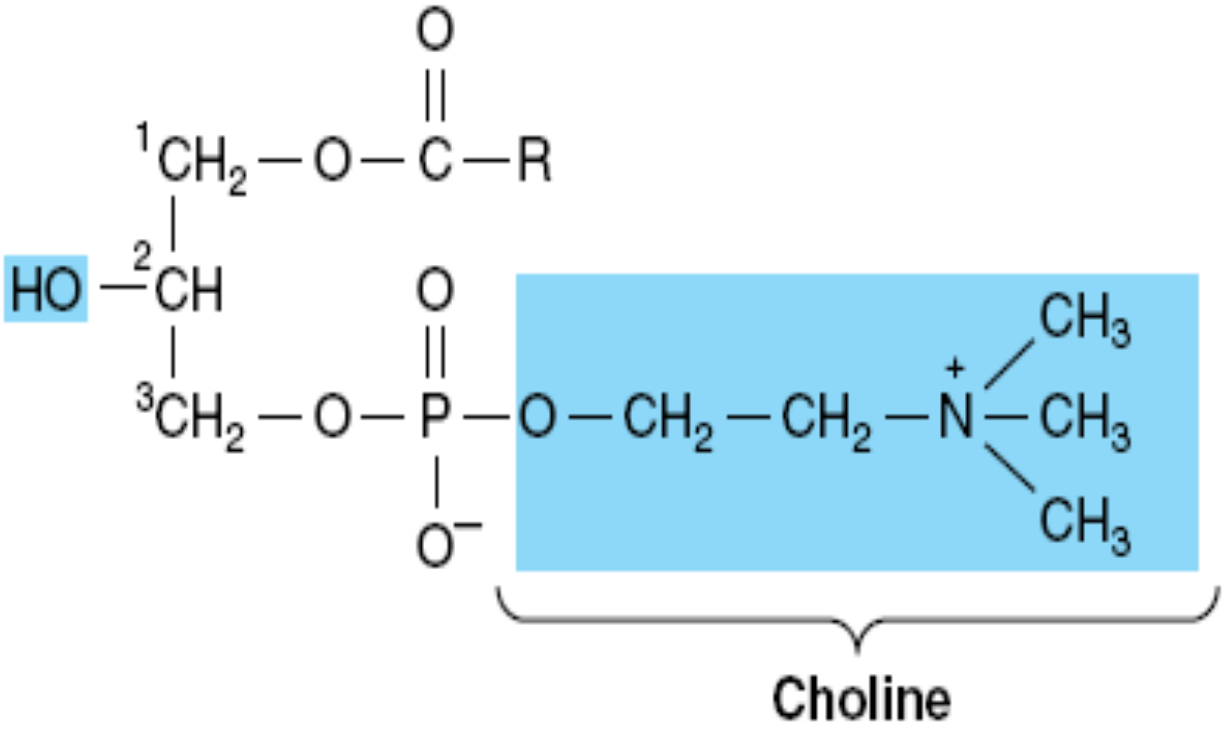




phosphatidylcholine



1,2-dihexadecanoyl-*sn*-glycero-3-phosphocholine



gure 14–9. Lysophosphatidylcholine (lysolecithin).

Phosphatidylcholine

- **If R1 & R2 are palmitic acid → Dipalmitoyl lecithin which is the major part of SURFACTANT.**

- **Surfactants reduce the surface tension of water by adsorbing at the liquid-gas interface**

- **Surfactant deficiency → RDS (Respiratory Distress Syndrome)**

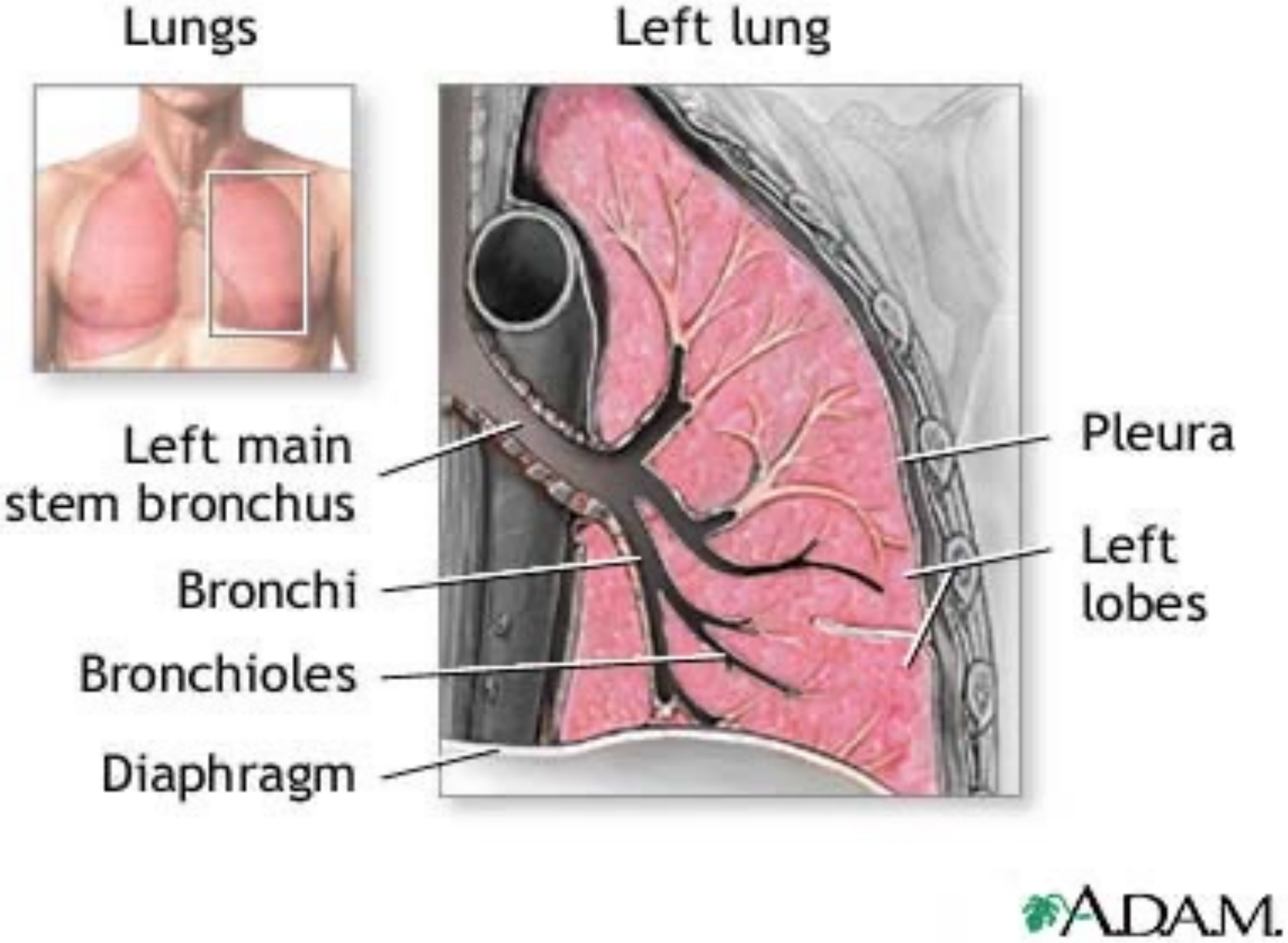
  In premature infants

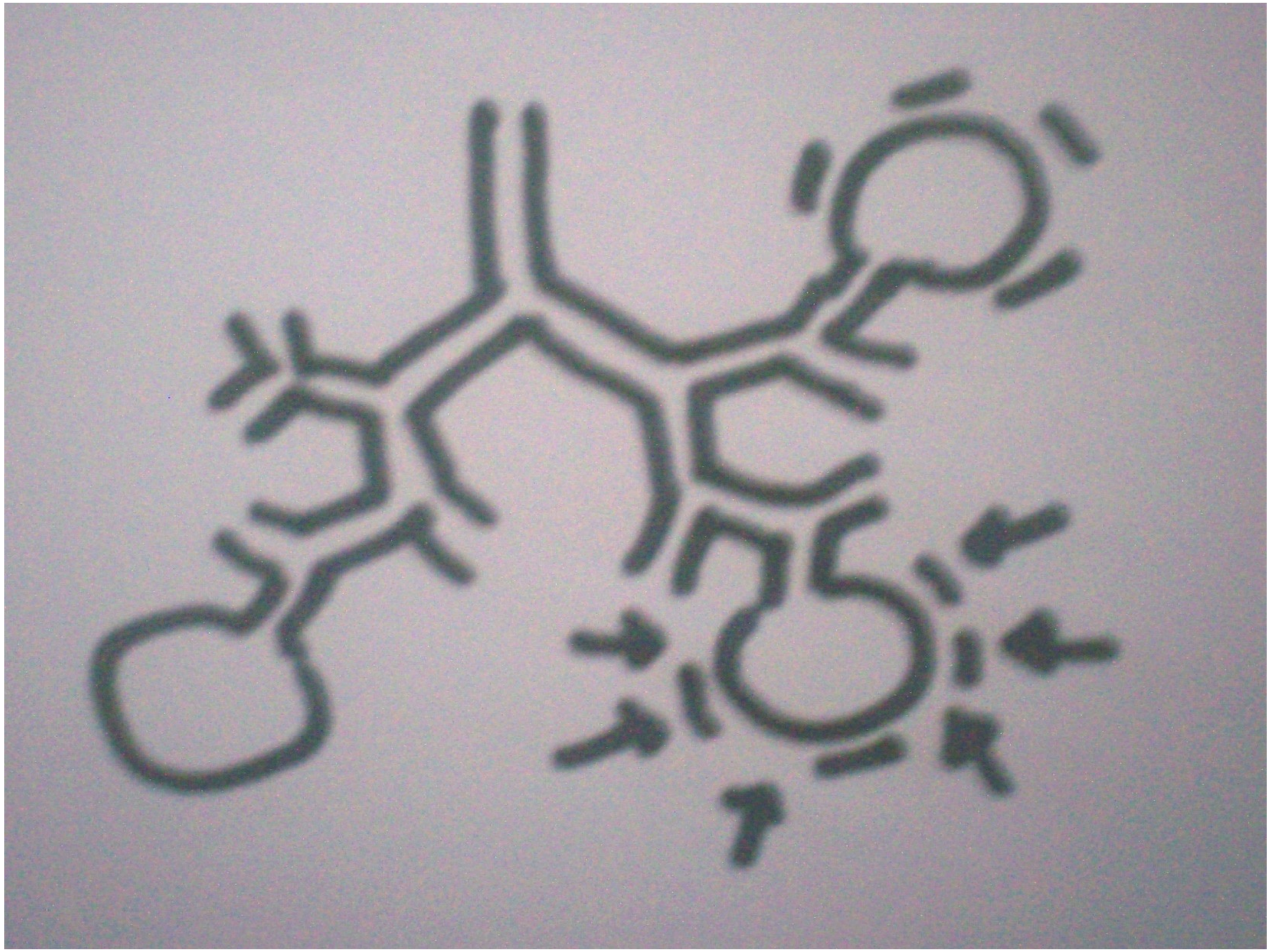
SURFACTANT

- **Adsorption is the adhesion of molecules of gas, liquid, or dissolved solids to a surface.[This process creates a film of the adsorbate (the molecules or atoms being accumulated) on the surface of the adsorbent.**

- **surfactant, in mammals has a relatively constant composition of about 90% lipid and 10% protein.**

- **surface tension reduction is due to dipalmitoylphosphatidylcholine (DPPC) which form about 50% of the total phospholipids in surfactant**


Lungs
Left lung
Left main
stem bronchus
Bronchi
Bronchioles
Diaphragm
Pleura
Left
lobes
A.D.A.M.

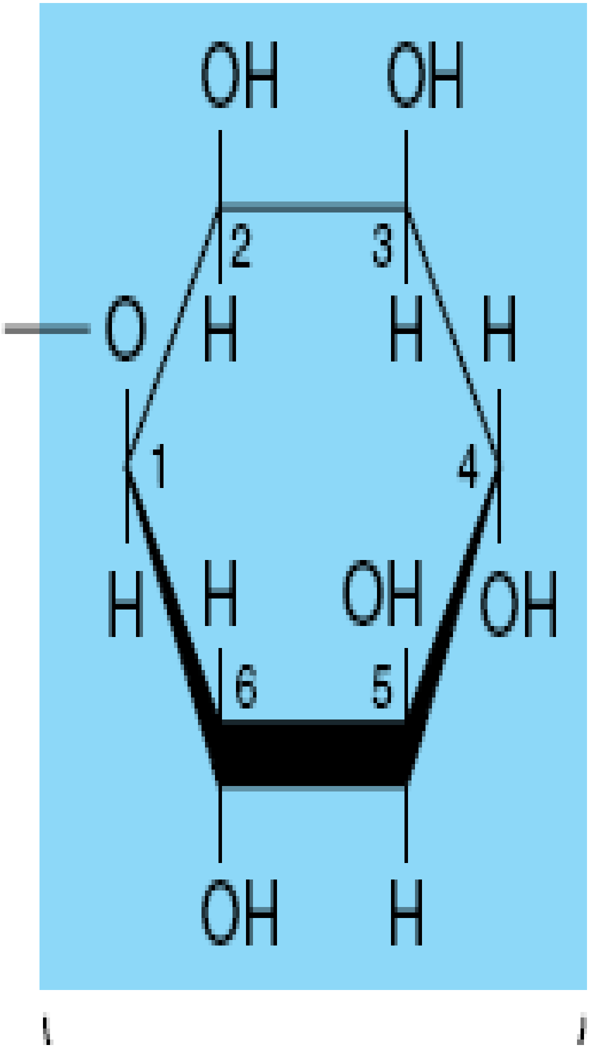


Phosphatidyl inositol

- **Phosphatidic acid + inositol:**

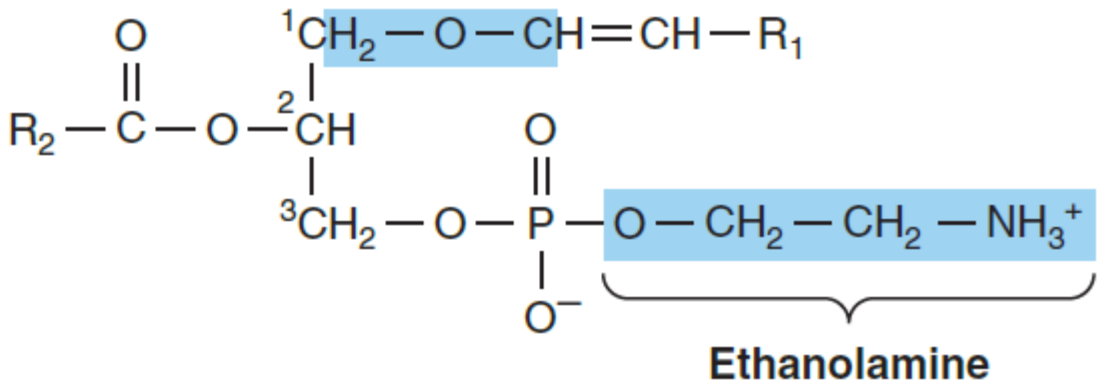
- **Source of second messengers:**

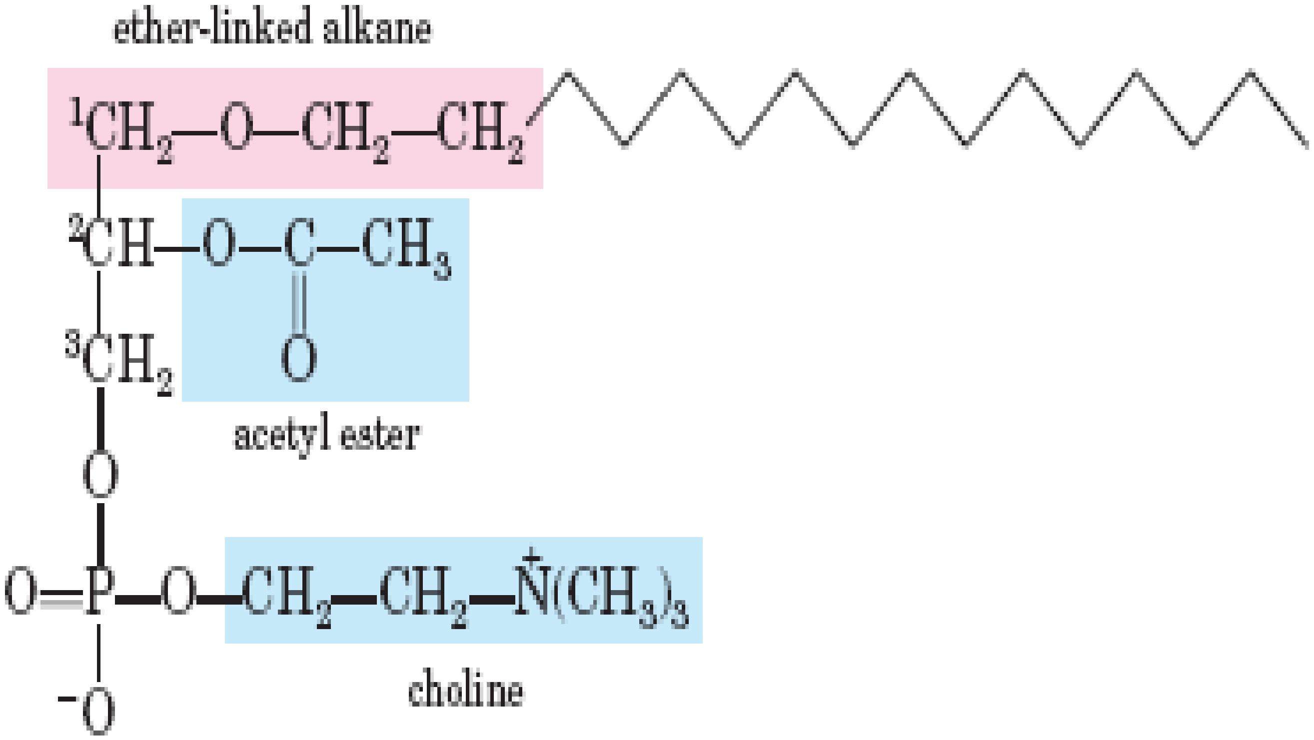
  IP3 & DAG.



Plasmalogens

- 10% to 30% of the phospholipids of brain and heart.
- resemble phosphatidylethanolamine but possess an ether link on the *sn-1 carbon*

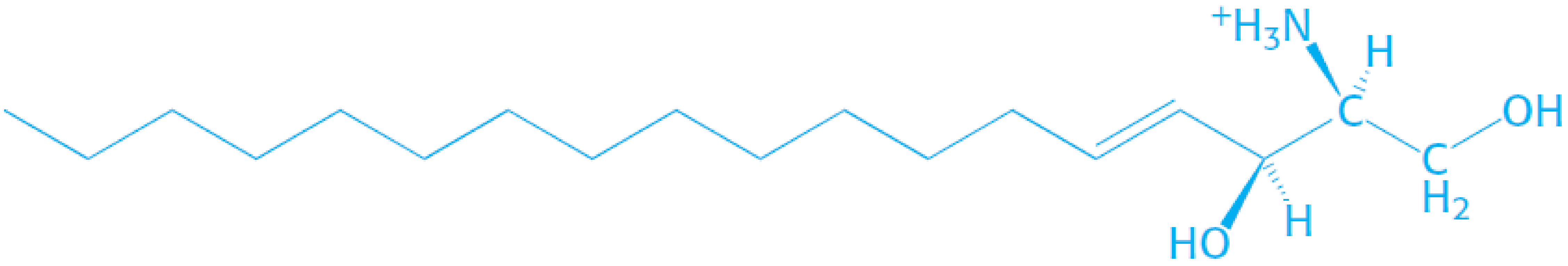




Platelet-activating factor

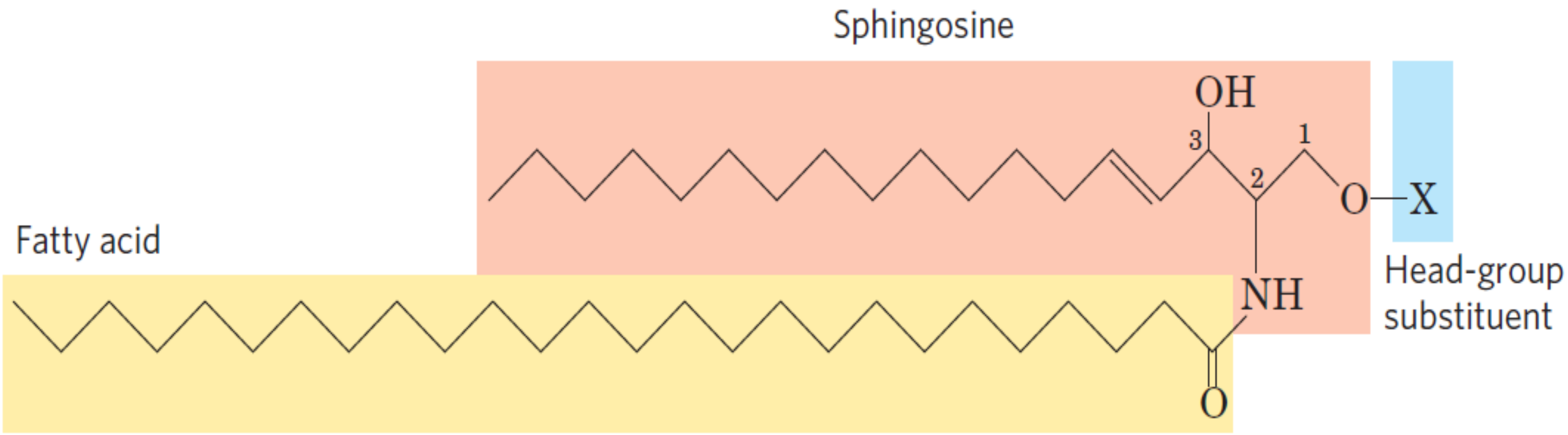
Sphingolipids

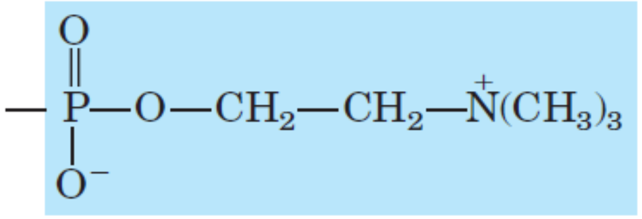
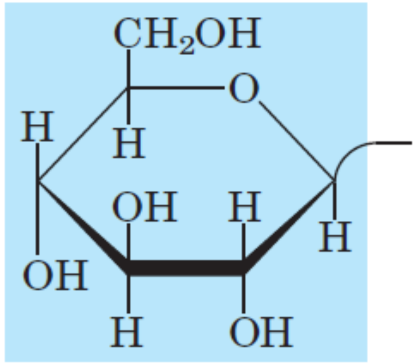
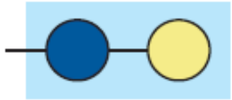
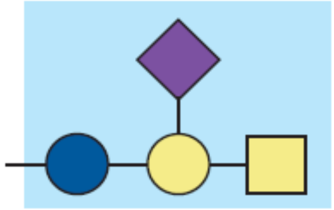
the phosphate is esterified to sphingosine, a complex amino alcohol



Sphingosine

Sphingolipids



Name of sphingolipid	Name of X—O	Formula of X
Ceramide	—	—H
Sphingomyelin	Phosphocholine	
Neutral glycolipids Glucosylcerebroside	Glucose	
Lactosylceramide (a globoside)	Di-, tri-, or tetrasaccharide	
Ganglioside GM2	Complex oligosaccharide	

Ceramide

- **When a fatty acid is attached in amide linkage to C-2.**
- **Sphingosine + fatty acid→ Ceramide**

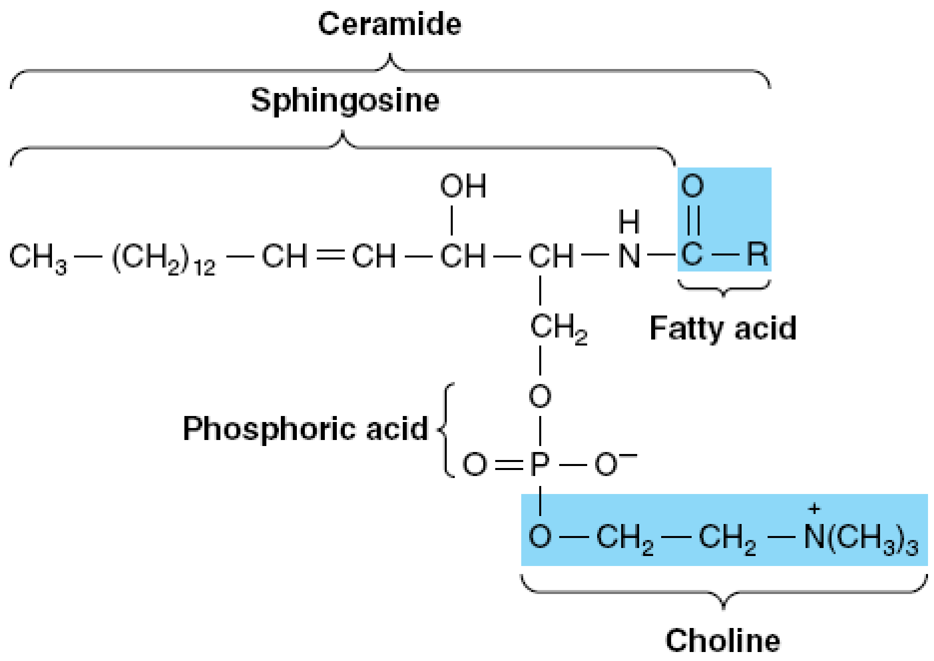
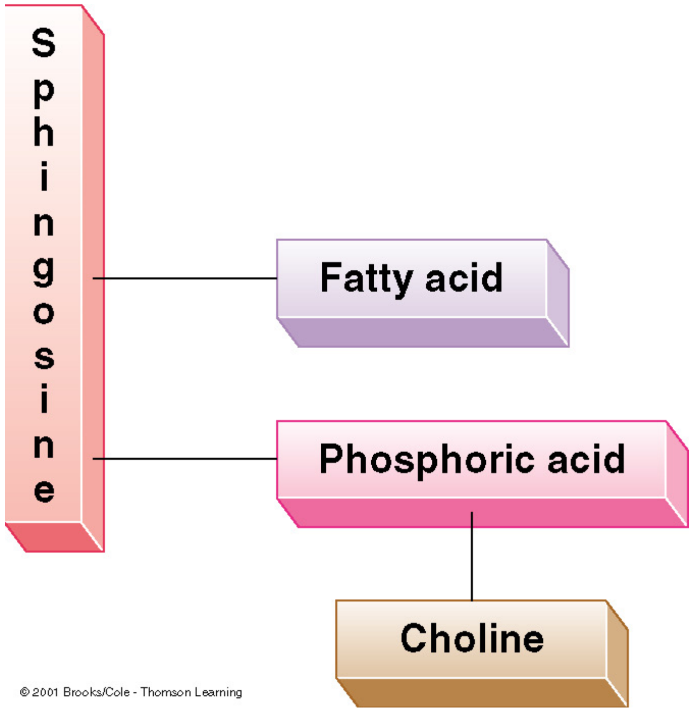


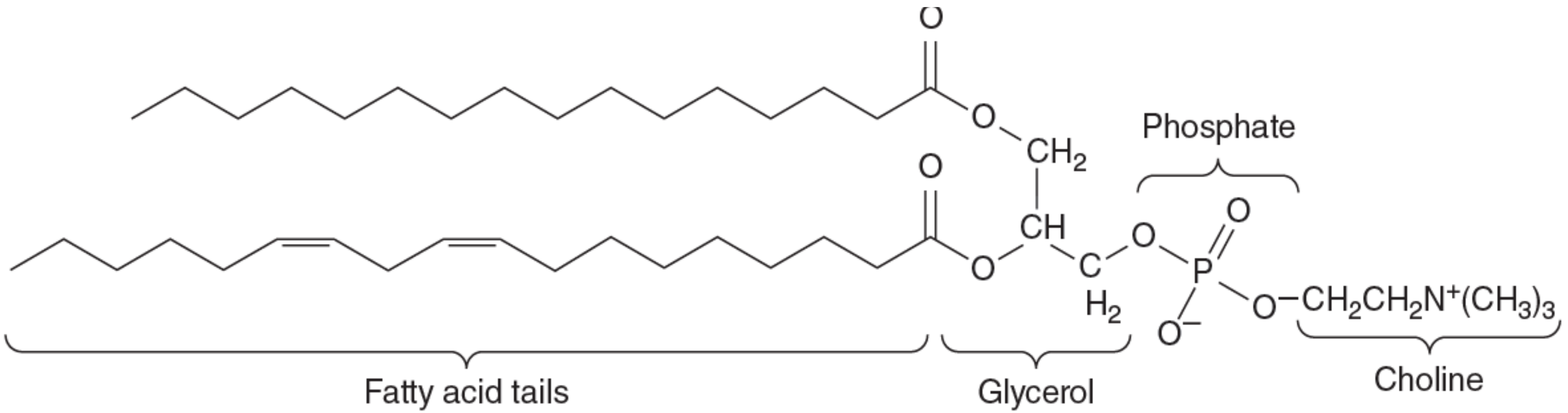
Figure 14–11. A sphingomyelin.

Sphingomyelin

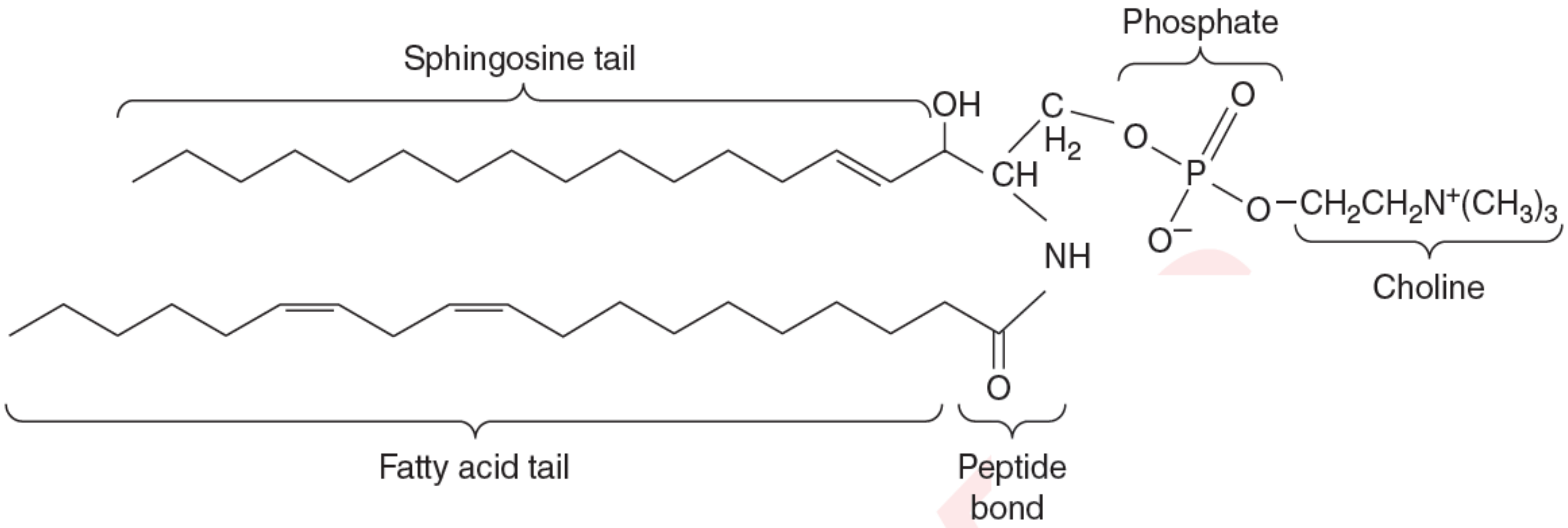
- sphingosine, fatty acid, phosphate and choline
- Sphingomyelins found in myelin sheath around neurons

A sphingomyelin.





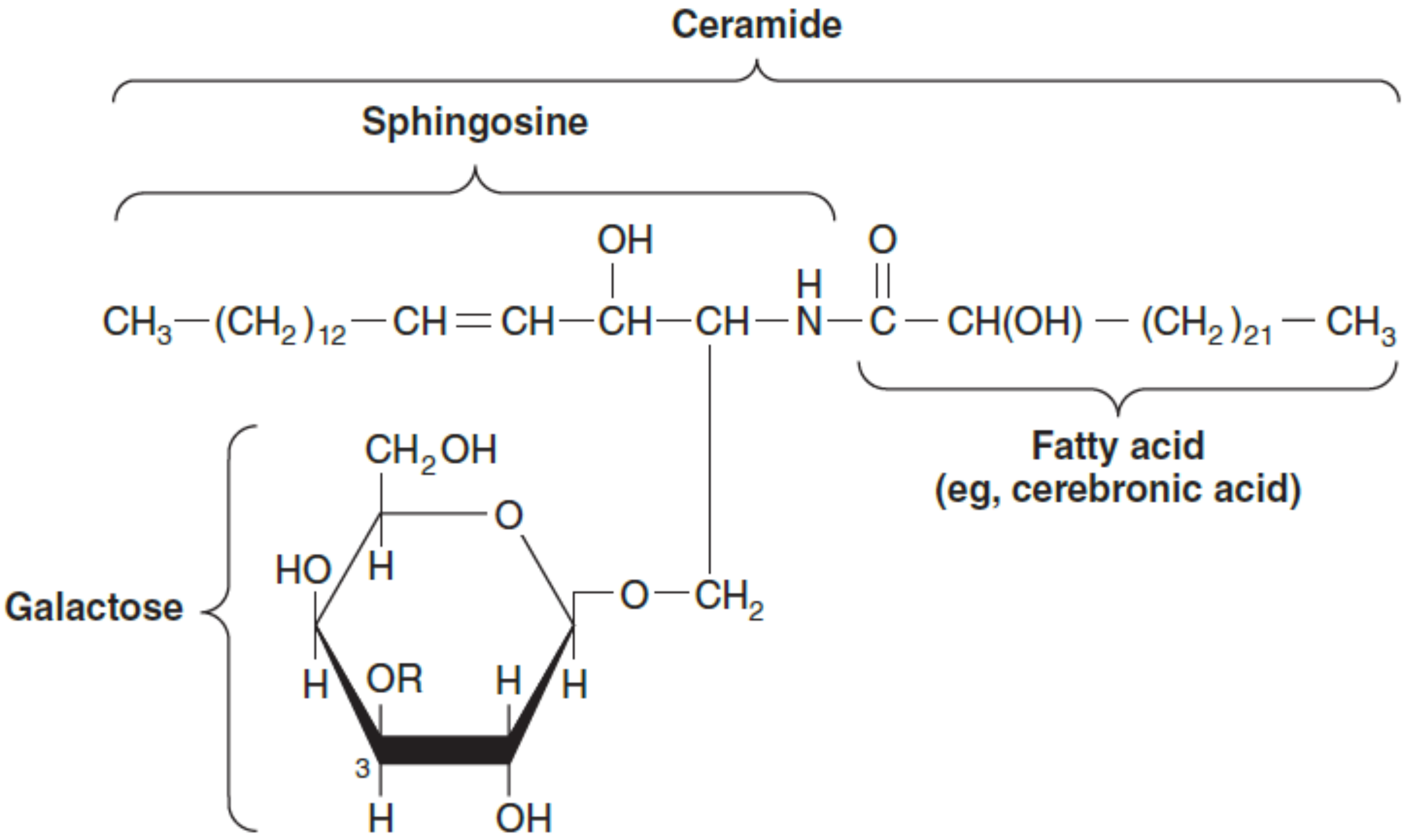
Phosphatidylcholine



A sphingomyelin

Glycolipids

- are lipids with an attached carbohydrate or carbohydrate chain. They are widely distributed in every tissue of the body, particularly in nervous tissue such as brain.



Structure of galactosylceramide.

Ceramide — Glucose — Galactose — *N*-Acetylgalactosamine
(Acyl-sphingo-sine)

Galactose: NeuAc

N-Acetylgalactosamine: Galactose

or

Cer — Glc — Gal — GalNAc — Gal

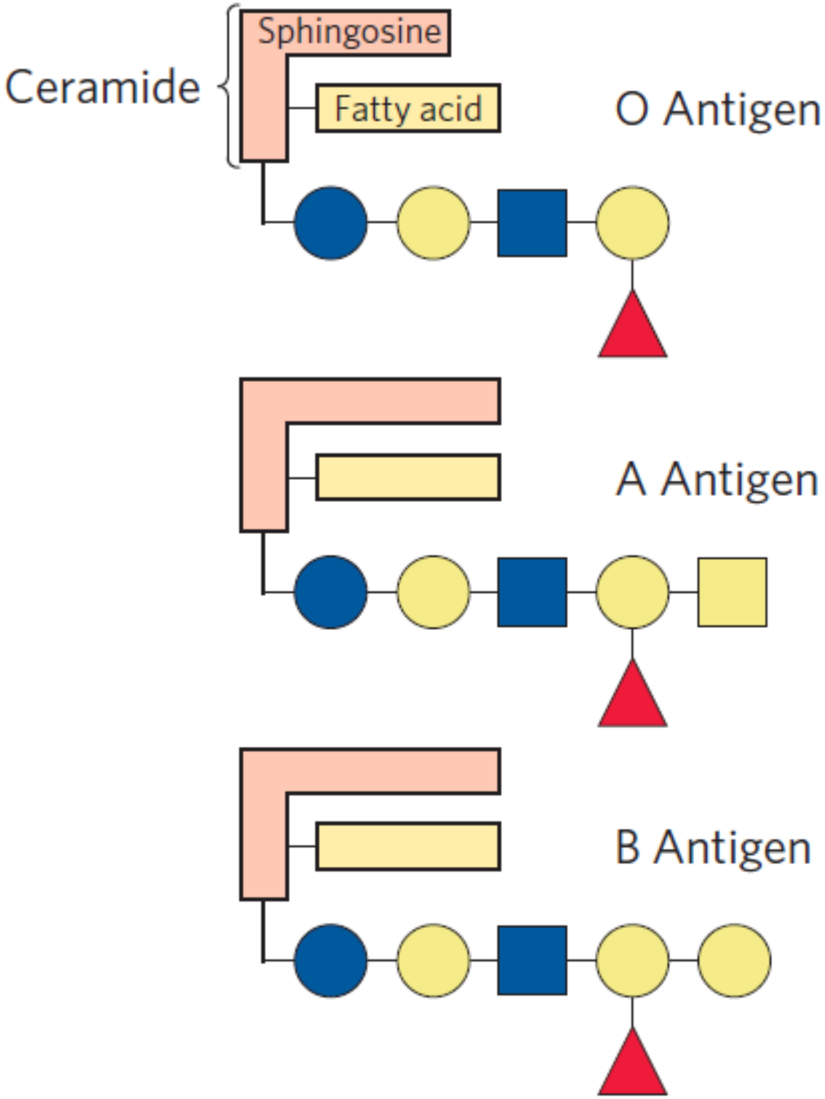
Gal: NeuAc

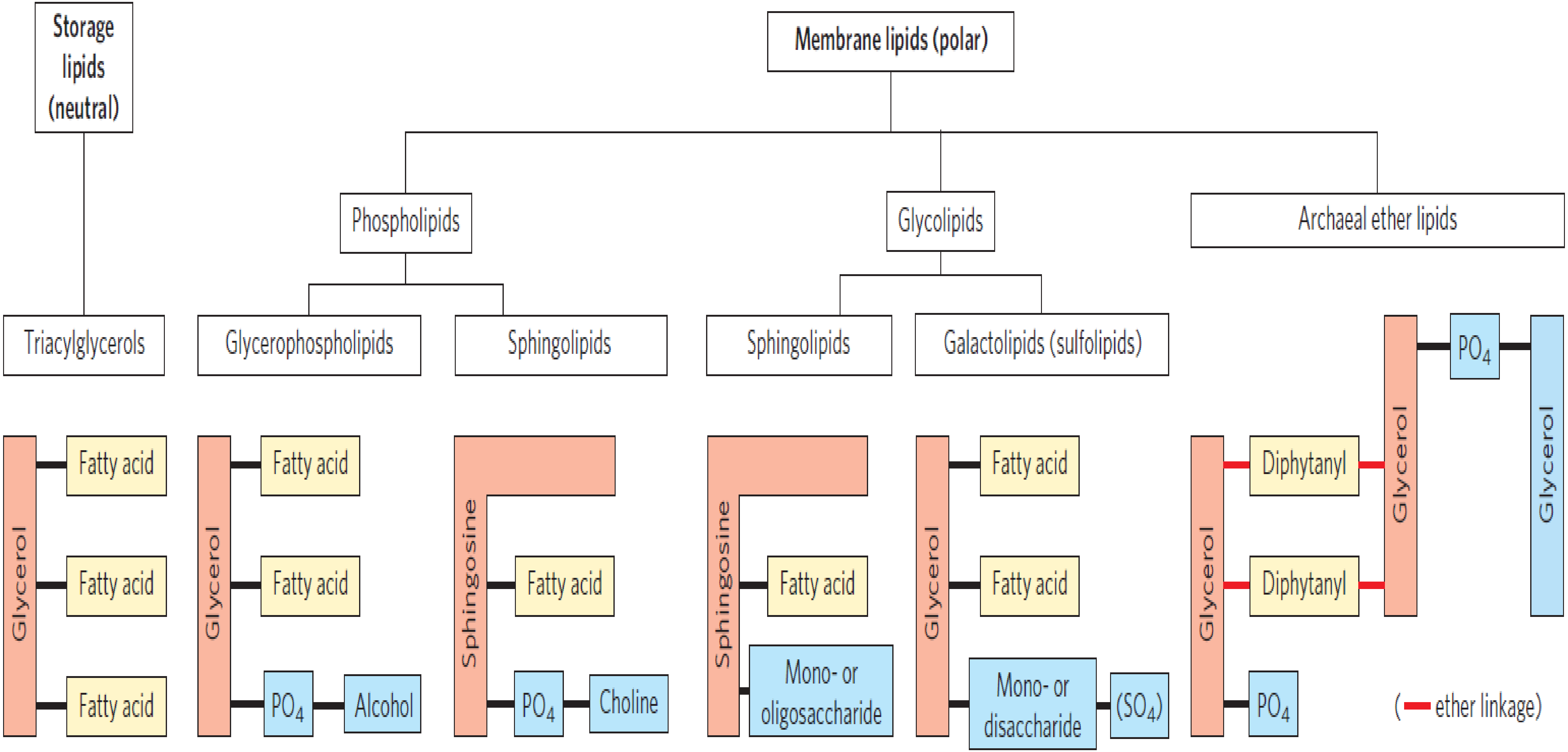
GM1 ganglioside, a monosialoganglioside, the receptor in human intestine for cholera toxin.

- **They are present on cell surfaces, with the oligosaccharides on the extracellular surface.**
- **They are found predominantly in the nervous system where they constitute 6% of all phospholipids.**
- **They are involved in various functions including cell-to-cell contact, ion conductance, and acting as receptors.**

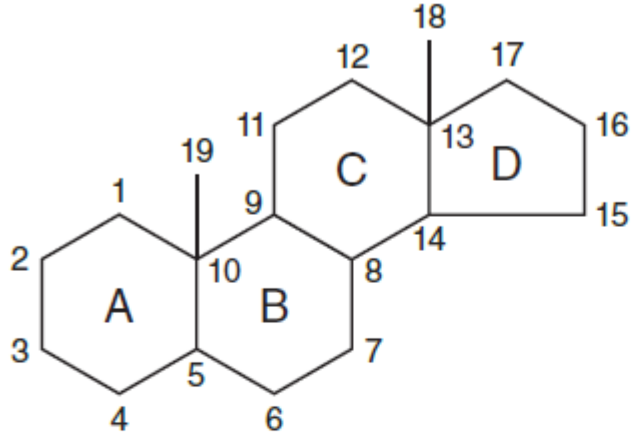
- **Structures similar to the ABO blood group antigens on the surface of human cells can be oligosaccharide components of glycosphingolipids in addition to being linked to proteins to form glycoproteins**

ABO Groups


Ceramide
Sphingosine
Fatty acid
O Antigen
A Antigen
B Antigen



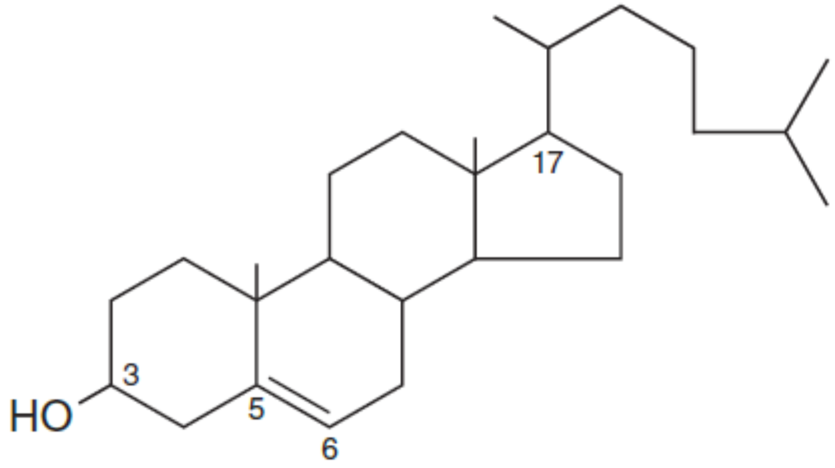
Storage lipids (neutral)
Membrane lipids (polar)
Phospholipids
Glycolipids
Archaeal ether lipids
Triacylglycerols
Glycerophospholipids
Sphingolipids
Sphingolipids
Galactolipids (sulfolipids)
Glycerol
Fatty acid
Fatty acid
Fatty acid
Glycerol
Fatty acid
Fatty acid
PO4
Alcohol
Sphingosine
Fatty acid
PO4
Choline
Sphingosine
Fatty acid
Mono- or oligosaccharide
Glycerol
Fatty acid
Fatty acid
Mono- or disaccharide
(SO4)
Glycerol
Diphytanyl
Diphytanyl
PO4
Glycerol
PO4
Glycerol
(ether linkage)



The steroid nucleus.

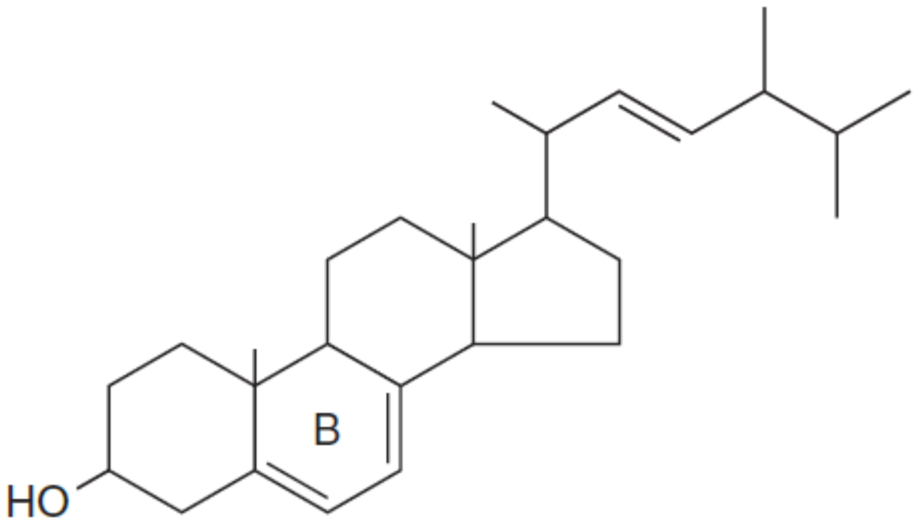
Cholesterol

- Association with atherosclerosis and heart disease
- is the precursor of a large number of equally important steroids that include the bile acids, adrenocortical hormones, sex hormones, vitamin D, and cardiac glycosides.
- is a major constituent of the plasma membrane and of plasma lipoproteins

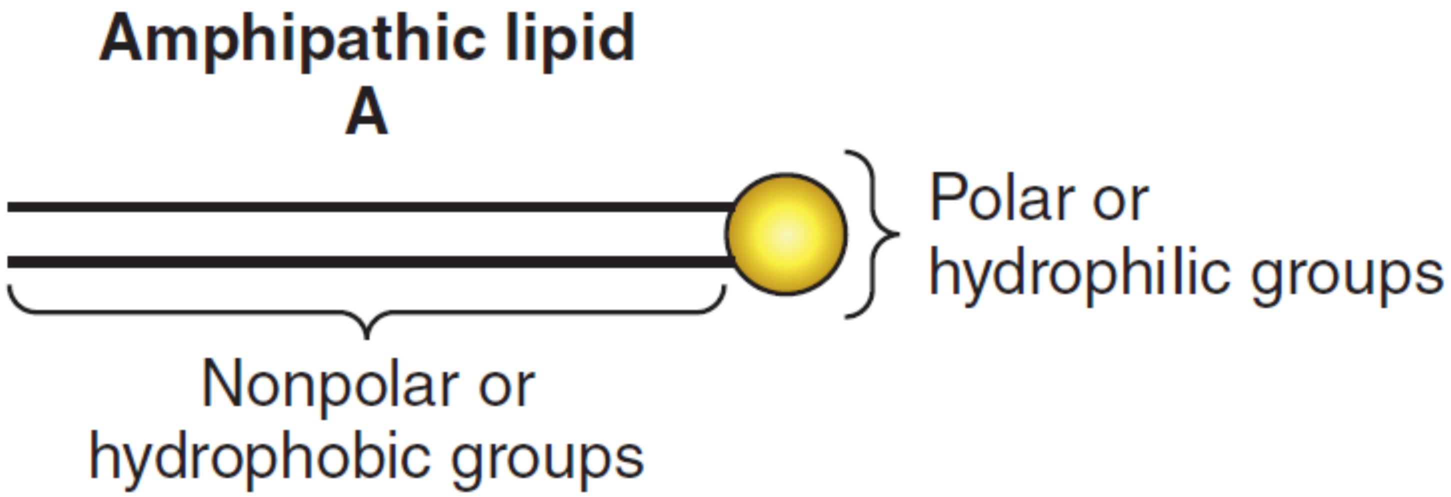


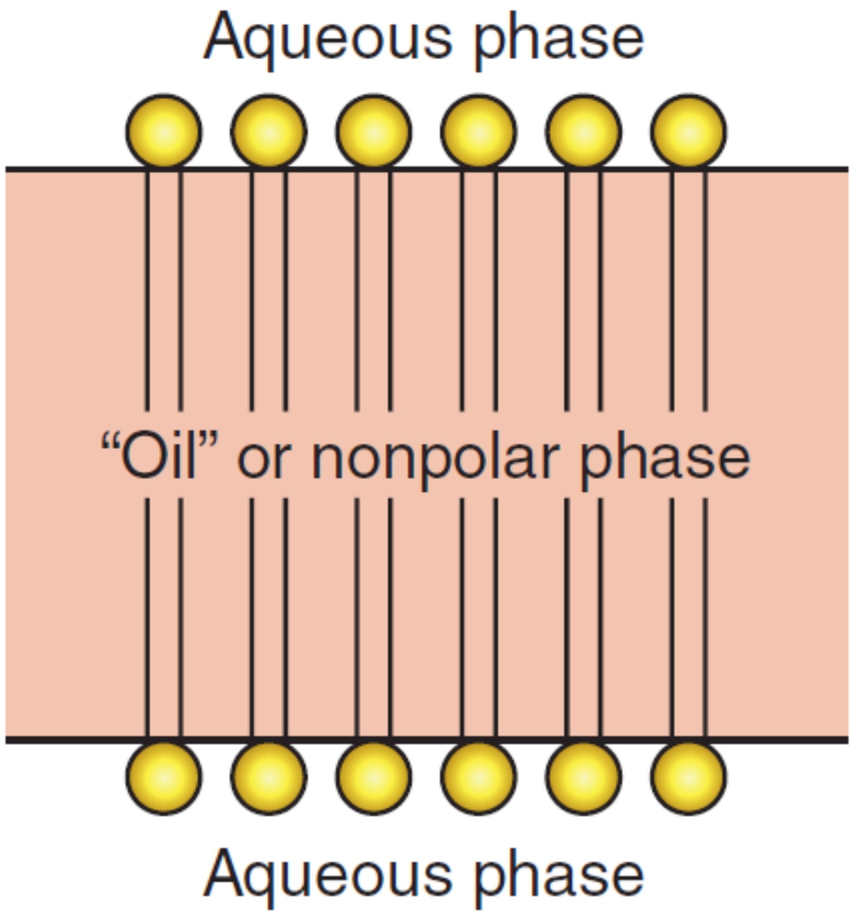
Cholesterol.

Vitamin D



Ergosterol.


Amphipathic lipid
A
Polar or
hydrophilic groups
Nonpolar or
hydrophobic groups



Lipid bilayer
B